# Exit Ticket · reflect and retrieve

## Exit Ticket

Every recursive call uses the same local-variable storage.

## Use key vocabulary from memory

1. Explain stack frame in your own words.

2. Explain unwind in your own words.

3. Explain local variable in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 182 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Recursion: call stack and tracing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Factorial(n) returns 1 when n=0; otherwise it returns n*Factorial(n-1). Trace the successive argument values in Factorial(3), stopping when the base case is reached.

### Q2

What is the difference between a function calling another function and returning a value to its caller?

### Q3

A stack contains bottom to top P, Q, R. After one pop, what is at the top?

### Q4

A function Square(x) returns x*x. Find Square(2)+Square(3), showing both calls.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Build the stack — SumTo(4) waits for SumTo(3), which waits for SumTo(2), which waits for SumTo(1).

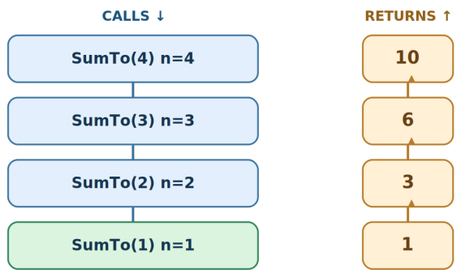
## Core idea 2

Separate the local values — Each stack frame holds its own n: 4, 3, 2 or 1. The values do not overwrite one another.

## Core idea 3

Unwind from the top — SumTo(1) returns 1; SumTo(2) returns 3; SumTo(3) returns 6; SumTo(4) returns 10.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

A pupil writes that SumTo(4) returns 4+3+2+1 before SumTo(1) has executed. Explain the order error.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

List the call-stack frames from bottom to top for SumTo(4).

## Question 2

Trace the returned values for SumTo(6).

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

stack frame — An entry containing the state for one active function call.

unwind — Return from calls in reverse order after a base case is reached.

local variable — A variable owned by a particular invocation.

call stack — Last-in, first-out storage for active calls.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

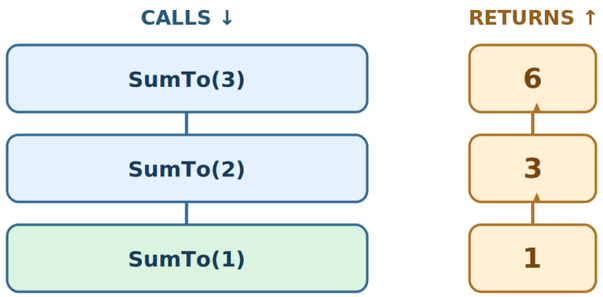
## Problem

Draw the call-stack frames for SumTo(3). Which call returns first?

## Follow the worked method

1. Push — The frames are SumTo(3), SumTo(2), then SumTo(1), with SumTo(1) at the top.
2. Stop — SumTo(1) is the base case; it returns 1 first.
3. Unwind — SumTo(2)=2+1=3, followed by SumTo(3)=3+3=6.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Call sequence
SumTo(5) → SumTo(4) → SumTo(3) → SumTo(2) → SumTo(1).

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.